

Practice Problems Sheet II

(Regular Languages)

Q1 Use Pumping Lemma to show that the following languages are not regular.

- (a) $\{x \in \{0, 1\}^* \mid x = x^R\}$
- (b) $\{x \in \{(,)\}^* \mid x \text{ is a balanced parenthesis}\}$
- (c) $\{0^n \mid n \text{ is a perfect cube}\}$
- (d) $\{x \in \{0, 1\}^* \mid |x| \text{ is a perfect square}\}$
- (e) $\{0^i 1^j \mid \gcd(i, j) = 1\}$

Q2

- (a) Prove the following generalization of the Pumping Lemma. A language L is regular iff there is a $n \in \mathbb{N}$ s.t. each $x = yzw \in L$, $|z| \geq n$, may be written as $x = yz_1z_2z_3w$ s.t. $|z_1z_2| \leq n$, $|z_2| \geq 1$ and for all integers $i \geq 0$, $yz_1z_2^i z_3w \in L$.
- (b) Use above generalization to show that language $L = \{01^i 2^i \mid i \geq 1\}$ is not regular. What is the difficulty in proving L to be non-regular using the original Pumping Lemma?

Q3 Show that language

$$L_{\text{prime}} = \{x \in \{0, 1\}^* \mid x \text{ is binary representation of a prime number}\}$$

is not regular.

(Hint: Use Pumping Lemma. You need to reason *mod* p , where p is a prime.)

Q4 A binary relation R on Σ^* is said to be *right invariant* iff for all $x, y \in \Sigma^*$ and $z \in \Sigma^*$ $[xRy \Rightarrow xzRyz]$.

- (a) Show that for any language L , relation R_L is right invariant.
- (b) Show that a language L is regular iff L is a union of some equivalence classes of a right invariant equivalence relation of a finite index.

Q5 In deciding properties, such as emptiness or finiteness, of regular languages in class we assumed that input regular language L is given as the transition graph of a DFA recognizing L . Suppose instead we use a different input model for L in which a DFA recognizing L is given as a black box. We are told the number of states n in this DFA and for any $x \in \Sigma^*$, we can run the DFA on x and observe the final result, that is, if x is accepted or rejected by the DFA. For this model solve the following.

- (a) Give an algorithm to decide if L is empty.
- (b) Give an algorithm to decide if L is infinite.
- (c) Give an algorithm to infer transition diagram of the minimum size DFA which recognizes L .

Q6 (Derivatives of a Regular Expression)

For a language L , over Σ and $a \in \Sigma$, define (another language)

$$\frac{dL}{da} = \{w \in \Sigma^* \mid aw \in L\}.$$

- (a) Show that derivative of a regular language is regular.

$$\left(\frac{dL}{da} \text{ is also written as } a \backslash L \right)$$

- (b) For a regular expression R , we write $\frac{dR}{da}$ to mean $\frac{dL(R)}{da}$.

Show that any regular expressions R, S ,

$$\frac{d(R + S)}{da} = \frac{dR}{da} + \frac{dS}{da}$$

- (c) Discover the expression for $\frac{d(RS)}{da}$ in the same spirit as in (b).

[Hint: Consider two cases, $\epsilon \in R$ and $\epsilon \notin R$]

- (d) What is the expression for $\frac{d(R^*)}{da}$?

(e) Compute the derivatives $\frac{d}{d0}$ and $\frac{d}{d1}$ for regular expressions $(0 + 1)^*110(0 + 1)^*$.

(f)[**optional**] Can you see how one may construct a DFA to recognize $L(R)$, using the idea of derivatives of R ?

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